

github-skills-challenge/src/aio: X AIOPS MSE Question Paper - at: X aiops_pipeline.py - github-skills: X + Ask Gemini

crispy-spoon-5ggw6wxr6p7jh4g65.github.dev

github-skills-challenge [Codespaces: crispy spoon]

Explorer

- github-skills-challenge [Cod...]
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- .devcontainer
- .github
- .pytest_cache
- data
- src
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 - aiops_pipeline.py
 - anomaly_detector.py
 - calculations.py
 - event_consumer.py
 - event_producer.py
 - event_topic.py
- tests
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 - calculations_test.py
 - test_aiops_pipeline.py
- .coveragerc
- .gitignore
- LICENSE
- pytest.ini
- README.md
- requirements.txt

Terminal

```
@jackstealer → /workspaces/github-skills-challenge (main) $ python - <<'PV'
import json
with open('data/service_data.json') as f:
    data = json.load(f)
print(data)
PV
[{'timestamp': '2026-09-2018:00:00', 'service': 'payment-service', 'response_time_ms': 120, 'cpu_percent': 42, 'memory_percent': 51, 'log_level': 'INFO', 'message': 'Payment request processed successfully'}, {'timestamp': '2026-09-2018:01:00', 'service': 'payment-service', 'response_time_ms': 135, 'cpu_percent': 45, 'memory_percent': 53, 'log_level': 'INFO', 'message': 'Payment request processed successfully'}, {'timestamp': '2026-09-2018:02:00', 'service': 'payment-service', 'response_time_ms': 128, 'cpu_percent': 44, 'memory_percent': 52, 'log_level': 'INFO', 'message': 'Payment request processed successfully'}, {'timestamp': '2026-09-2018:03:00', 'service': 'payment-service', 'response_time_ms': 142, 'cpu_percent': 48, 'memory_percent': 55, 'log_level': 'INFO', 'message': 'Payment request processed successfully'}, {'timestamp': '2026-09-2018:04:00', 'service': 'payment-service', 'response_time_ms': 130, 'cpu_percent': 46, 'memory_percent': 54, 'log_level': 'INFO', 'message': 'Payment request processed successfully'}, {'timestamp': '2026-09-2018:05:00', 'service': 'payment-service', 'response_time_ms': 610, 'cpu_percent': 75, 'memory_percent': 70, 'log_level': 'ERROR', 'message': 'Payment service timeout'}, {'timestamp': '2026-09-2018:06:00', 'service': 'payment-service', 'response_time_ms': 640, 'cpu_percent': 94, 'memory_percent': 91, 'log_level': 'ERROR', 'message': 'Database connection timeout'}, {'timestamp': '2026-09-2018:07:00', 'service': 'payment-service', 'response_time_ms': 150, 'cpu_percent': 49, 'memory_percent': 56, 'log_level': 'INFO', 'message': 'Payment request processed successfully'}, {'timestamp': '2026-09-2018:08:00', 'service': 'payment-service', 'response_time_ms': 138, 'cpu_percent': 47, 'memory_percent': 55, 'log_level': 'INFO', 'message': 'Payment request processed successfully'}, {'timestamp': '2026-09-2018:09:00', 'service': 'payment-service', 'response_time_ms': 145, 'cpu_percent': 50, 'memory_percent': 57, 'log_level': 'INFO', 'message': 'Payment request processed successfully'}]
@jackstealer → /workspaces/github-skills-challenge (main) $ python - <<'PV'
import json
with open('data/service_data.json') as f:
    data = json.load(f)
for r in data:
    if r['response_time_ms'] > 500 or r['cpu_percent'] > 80 or r['memory_percent'] > 80 or r['log_level'] == 'ERROR':
        print('ANOMALY:', r)
PV
ANOMALY: {'timestamp': '2026-09-2018:05:00', 'service': 'payment-service', 'response_time_ms': 610, 'cpu_percent': 75, 'memory_percent': 70, 'log_level': 'ERROR', 'message': 'Payment service timeout'}
ANOMALY: {'timestamp': '2026-09-2018:06:00', 'service': 'payment-service', 'response_time_ms': 640, 'cpu_percent': 94, 'memory_percent': 91, 'log_level': 'ERROR', 'message': 'Database connection timeout'}
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ANOMALY: {'timestamp': '2026-09-2018:05:00', 'service': 'payment-service', 'response_time_ms': 610, 'cpu_percent': 75, 'memory_percent': 70, 'log_level': 'ERROR', 'message': 'Payment service timeout'}
@jackstealer → /workspaces/github-skills-challenge (main) $ python - <<'PV'
import json
with open('data/service_data.json') as f:
    data = json.load(f)
for r in data:
    import json
    with open('data/service_data.json') as f:
        data = json.load(f)
    d = AnomalyDetector()
    for r in data:
        e = d.detect(r)
        if e:
            print(e)
PV
{'timestamp': '2026-09-2018:05:00', 'service': 'payment-service', 'type': 'ANOMALY', 'reasons': ['High response time'], 'source': {'timestamp': '2026-09-2018:05:00', 'service': 'payment-service', 'response_time_ms': 610, 'cpu_percent': 75, 'memory_percent': 70, 'log_level': 'ERROR', 'message': 'Payment service timeout'}}
{'timestamp': '2026-09-2018:06:00', 'service': 'payment-service', 'type': 'ANOMALY', 'reasons': ['High response time', 'High CPU utilization', 'High memory utilization'], 'source': {'timestamp': '2026-09-2018:06:00', 'service': 'payment-service', 'response_time_ms': 640, 'cpu_percent': 94, 'memory_percent': 91, 'log_level': 'ERROR', 'message': 'Database connection timeout'}}
@jackstealer → /workspaces/github-skills-challenge (main) $ python - <<'PV'
from src.event_topic import EventTopic
from src.event_producer import EventProducer
from src.anomaly_detector import AnomalyDetector
import json
with open('data/service_data.json') as f:
    data = json.load(f)
topic = EventTopic('service-events')
producer = EventProducer(topic)
d = AnomalyDetector()
for r in data:
    e = d.detect(r)
    if e:
        producer.publish(e)
print('Messages in topic:', len(topic.get_messages()))
print(topic.get_messages())
```

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Problems Output Debug Console Terminal Ports

```
python --help
python -m src.aiops_pipeline
python -m src.event_producer
python -m src.event_consumer
python -m src.anomaly_detector
python -m src.event_topic
python -m src.calculations
python -m src.coverage
python -m src.gitignore
python -m src.LICENSE
python -m src.pytest.ini
python -m src.README.md
python -m src.requirements.txt
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- Timeline

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```
topic = EventTopic('service-events')
new_kafka = EventProducer('main')
@jackstealer -> /workspaces/github-skills-challenge (main) $ python -c 'PR'
from src.event_topic import EventTopic
from src.event_producer import EventProducer
from src.event_consumer import EventConsumer
from src.anomaly_detector import AnomalyDetector
producer.publish(e)
print('Consumed:', consumer.consume())
py
Consumed: [{'timestamp': '2026-09-20T10:05:00', 'service': 'payment-service', 'type': 'ANOMALY', 'reasons': ['High response time'], 'source': {'timestamp': '2026-09-20T10:05:00', 'service': 'payment-service', 'response_time_ms': 610, 'cpu_percent': 75, 'memory_percent': 70, 'log_level': 'ERROR', 'message': 'Payment service timeout'}}, {'timestamp': '2026-09-20T10:06:00', 'service': 'payment-service', 'type': 'ANOMALY', 'reasons': ['High response time', 'High CPU utilization', 'High memory utilization'], 'source': {'timestamp': '2026-09-20T10:06:00', 'service': 'payment-service', 'response_time_ms': 640, 'cpu_percent': 94, 'memory_percent': 91, 'log_level': 'ERROR', 'message': 'Database connection timeout'}}]
@jackstealer -> /workspaces/github-skills-challenge (main) $ PYTHONPATH=. python -m src.aiops_pipeline
AIOPS Pipeline Result
=====
Records processed: 10
Anomalies detected: 2
Events consumed: 2
Detected Events:
Service: payment-service
Timestamp: 2026-09-20T10:05:00
Type: ANOMALY
Reasons: High response time
Service: payment-service
Timestamp: 2026-09-20T10:06:00
Type: ANOMALY
Reasons: High response time, High CPU utilization, High memory utilization
@jackstealer -> /workspaces/github-skills-challenge (main) $ PYTHONPATH=. pytest -q
```

Codespaces: crispy spoon | main | Deborah (9 hours ago) | Ln 62, Col 32 | Spaces: 4 | UTF-8 | LF | Python | Python 3.13.5.final.0 | Layout: US

Type here to search | 31°C Sunny | 11:01 AM 9/21/2026

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Records processed: 10
Anomalies detected: 2
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Detected Events:
Service: payment-service
Timestamp: 2026-09-20T10:05:00
Type: ANOMALY
Reasons: High response time
Service: payment-service
Timestamp: 2026-09-20T10:06:00
Type: ANOMALY
Reasons: High response time, High CPU utilization, High memory utilization
@jackstealer -> /workspaces/github-skills-challenge (main) $ PYTHONPATH=. pytest -q
===== ERRORS =====
ERROR collecting tests/test_aiops_pipeline.py
ImportError while importing test module '/workspaces/github-skills-challenge/tests/test_aiops_pipeline.py'.
Hint: make sure your test modules/packages have valid Python names.
Traceback:
/usr/local/lib/python3.13/importlib/_init_.py:88: in import module
return _bootstrap._gcd_import(name[level:], package, level)
tests/test_aiops_pipeline.py:3: in <module>
from src.anomaly_detector import AnomalyDetector
E ModuleNotFoundError: No module named 'src'
===== short test summary info =====
ERROR tests/test_aiops_pipeline.py
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!! Interrupted: 1 error during collection !!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
1 error in 0.13s
@jackstealer -> /workspaces/github-skills-challenge (main) $ cd /workspaces/github-skills-challenge && pytest -q
8 passed in 0.02s [100%]
@jackstealer -> /workspaces/github-skills-challenge (main) $ []
```

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